

AI Career Intelligence Platform

Testing Strategy Document

Test Strategy • Performance Tests • Security Tests

Version: 1.0

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1. Test Strategy

The testing approach validates functionality, reliability, security and performance across the AI Career Intelligence Platform.

Test Type	Purpose
Unit Testing	Validate individual functions and services
Integration Testing	Verify module interactions
API Testing	Validate REST endpoints
UI Testing	Check frontend workflows
User Acceptance Testing	Confirm business requirements

2. Performance Testing

Performance testing ensures the application remains responsive under expected workloads.

Metric	Target
API Response Time	< 500 ms
Concurrent Users	500+
Resume Upload	< 5 seconds
Database Queries	Optimized with indexes
Availability	99.9% uptime target

- Use Apache JMeter or k6 for load testing.
- Monitor CPU, memory and database performance.
- Identify bottlenecks before production deployment.

3. Security Testing

Security Test	Objective
Authentication	Validate JWT and session security
Authorization	Verify role-based access
SQL Injection	Ensure parameterized queries
Cross-Site Scripting (XSS)	Validate input sanitization
File Upload	Check malicious file protection
Dependency Scan	Identify vulnerable packages

Recommended tools include OWASP ZAP, SonarQube, Bandit, and GitHub Dependabot.

4. Conclusion

This document defines the testing strategy for Version 1.0 of the AI Career Intelligence Platform. It establishes quality assurance practices covering functional testing, performance validation and security verification before production release.